Document Control

Title **Antibiotic guidelines for Gastroenterological indications in Secondary Care: Appendicitis and other intra-abdominal infection, including perforation**			
Author			**Author's job title** Consultant Microbiologist Antibiotic Pharmacist Consultant Surgeon
Directorate Diagnostics			**Department** Pathology
Version	**Date Issued**	**Status**	**Comment / Changes / Approval**
0.1	Oct 2015	Draft	Supercedes Antibiotic guidelines for Gastroenterological indications in Secondary Care v1.5. Guidance split by indication. Inserted into new trust template. Circulated to AWG.
0.2Feb	Feb 2016	Draft	Review of UpToDate. No significant changes. Change of format to match RxGuidelines app
1.0	Mar 2016	Final	Ratified at Drug and Therapeutics Committee 17/3/16 Minor change to gentamicin link and oral switch guidance
1.1	Nov 2019	Revision	Updated references. Updated approval process. Addition of diagnostic sign names at junior Drs suggestion to aid writing up results of examination. Separation of conservative management oral treatment option.
2.0	Feb 2020	Final	Feedback received from colorectal surgeons. Revised imaging, examination, pre-op advice, diagnostics, course length for conservative management. Approved by IPDG.
2.1	Nov 2020	Revision	Clarification of patient selection for non-operative vs operative management following advice from colorectal surgeons. Addition of paediatric guidance.
2.2	Jan 2021	Revision	Addition of NOTA and clinical scoring evidence review in associated documentation section, written by Consultant Surgeon
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Lead Director Director of Medicine			
Superseded Documents Antibiotic guidelines for Gastroenterological indications in Secondary Care: Appendicitis and other intra-abdominal infection, including perforation v1.2 06022020			
Issue Date Jan 2021	**Review Date** Dec 2023	**Review Cycle** Three years	

Consulted with the following stakeholders: • Antibiotic Working Group • Infection Prevention and Control Committee • Consultant Gastroenterologists • Consultant General Surgeons	
Approval and Review Process • Clinical Audit and Guidelines Group	
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Policy categories for Trust's internal website (Bob) Pharmacy, microbiology	**Tags for Trust's internal website (Bob)** Appendicitis, perforation, abdominal sepsis

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1. Purpose

1.1. This document sets out Northern Devon Healthcare NHS Trust's best practice guidelines for appropriate microbiological investigation and antimicrobial prescribing in adult and paediatric patients with appendicitis and intra-abdominal infection.

1.2. This guideline applies to all adult and paediatric patients and must be adhered to. Special considerations exist for pregnant and breastfeeding patients; liaise with specialist clinicians as appropriate in these cases.

1.3. Non-compliance with this guideline may be for valid clinical reasons only. The reason(s) for non-compliance must be documented clearly in the patient's notes.

1.4. This guideline is primarily aimed at all prescribing teams but other staff (e.g. nursing staff, pharmacists) may need to familiarise themselves with some aspects of the guideline.

1.5. Implementation of this guideline will ensure that infections in this guidance are managed according to current evidence and standards of practice. A standard of care is specified to facilitate a consistent approach between ward-based clinical teams, microbiology and pharmacy in terms of patient management, specimen processing and drug availability.

2. Responsibilities

2.1. Responsibility for education and training on antibiotic use in NDHT lies with the Lead Consultant Microbiologist for Antibiotic Stewardship. It will be provided through formal study days and informal training on the ward.

2.2. The author will be responsible for ensuring the guidelines are reviewed and revisions approved by the Clinical Audit and Guidelines Group in accordance with the Document Control Report.

2.3. All versions of these guidelines will be archived in electronic format by the author within the Antibiotic Stewardship policy archive.

2.4. Minor changes to guidance can be made by the Lead Clinician for Antibiotic Stewardship without need for formal ratification. These changes will be documented in the document control report.

2.5. Any revisions to the final document will be recorded on the Document Control Report.

2.6. To obtain a copy of the archived guidelines, contact should be made with the author.

2.7. Monitoring of implementation, effectiveness and compliance with these guidelines will be the responsibility of the Lead Clinician for Antibiotic Stewardship. Where non-compliance is found, the reasons for this must have been documented in the patient's medical notes.

Role of Antibiotic Working Group (AWG)

2.8. The AWG is responsible for:

- Leading antibiotic guideline development and review within Northern Devon Healthcare Trust
- Involving all relevant stakeholders in guideline development and review

3. Contacts

3.1. Contact numbers:

- Microbiologist — Bleep 193. Via switchboard out of hours.
- Antibiotic Pharmacist — Bleep 029 (Mon-Fri only)

4. Management of Appendicitis and other intra-abdominal infections (including perforation)

- See appendix 1 for adult management
- See appendix 2 for paediatric management

5. Monitoring Compliance with and the Effectiveness of the Policy

Suggested audit criteria

5.1. The following could be used:

- Percentage of patients with sample sent
- Percentage of patients managed conservatively with antibiotics
- Percentage of patients initially managed conservatively who subsequently underwent surgery due to complications
- Percentage of patients who received appropriate peri-operative antibiotics

Process for Implementation and Monitoring Compliance and Effectiveness

5.2. Incidents involving gastroenterological infection should be reported according to the Trust's Incident Reporting Policy. Critical incident reports relating to gastroenterological infection will be collated by the Antibiotic Pharmacist. Results will be reported on an annual basis to the Clinical Effectiveness Committee.

6. Equality Impact Assessment

6.1. The author must include the Equality Impact Assessment Table and identify whether the policy has a positive or negative impact on any of the groups listed. The Author must make comment on how the policy makes this impact.

Table 1: Equality impact Assessment

Group	Positive Impact	Negative Impact	No Impact	**Comment**
Age	X			Separate guidance for adults and paediatrics
Disability			X	
Gender			X	
Gender Reassignment			X	
Human Rights (rights to privacy, dignity, liberty and non-degrading treatment)			X	
Marriage and civil partnership			X	
Pregnancy		X		Some treatment advice may harm the unborn foetus,

				discuss on a case-by-case basis with Obstetricians and Pharmacy for advice.
Maternity and Breastfeeding		X		Some treatments may be excreted in breast milk. Discuss on a case-by-case basis with Paediatricians and Pharmacy for advice.
Race (ethnic origin)			X	
Religion (or belief)			X	

7. References

- Solomkin JS et al. (2009) Diagnosis and Management of Complicated Intra-abdominal Infection in Adults and Children: Guidelines by the Surgical Infection Society and the Infectious Diseases Society of America Clin Infect Dis 50, 133-64
- http://jama.jamanetwork.com/article.aspx?articleid=2320315 Salminen P, Paajanen H, Rautio T, Nordström P, Aarnio M, Rantanen T, Tuominen R, Hurme S, Virtanen J, Mecklin JP, Sand J, Jartti A, Rinta-Kiikka I, Grönroos JM. 2015. Antibiotic Therapy vs Appendectomy for Treatment of Uncomplicated Acute Appendicitis: The APPAC Randomized Clinical Trial. JAMA 313:23;2340-8
    - Trialled surgical intervention vs antibiotics for uncomplicated appendicitis on CT. 10 days abx therapy if no surgery
    - Recurrence of appendicitis requiring surgical intervention within 12 months in 70 out of 257 patients, but no increased complications from delaying appendectomy.
- http://www.nejm.org/doi/full/10.1056/NEJMoa1411162#t=articleTop Sawyer, RG; Claridge, JA; Nathens, AB; Rotstein, OD; Duane, TM; Evans, HL; Cook, CH; O'Neill, PJ; Mazuski, JE; Askari, R; Wilson, MA; Napolitano, LM; Namias, N; Miller, PR; Dellinger, EP; Watson, CM; Coimbra, R; Dent, DL; Lowry, SF; Cocanour, CS; West, MA; Banton, KL; Cheadle, WG; Lipsett, PA; Guidry, CA; Popovsky, K. 2015. Trial of Short-Course Antimicrobial Therapy for Intra-abdominal Infection. N Engl J Med 2015; 372:1996-2005 DOI: 10.1056/NEJMoa1411162
    - Supports short courses (3-5 days max) once source control achieved surgically
    - STOP-IT trial
    - Recurrence of disease likely due to inadequate source control in original infection
- SIGN 2014. SIGN Guideline 104: Antibiotic Prophylaxis in Surgery. http://www.sign.ac.uk/pdf/sign104.pdf
    - Supports giving antibiotic prophylaxis prior to appendicectomy

- Smink, D. 2018. Management of Acute Appenditicitis in Adults. Topic 1383 Version 55.0 via Up To Date https://www.uptodate.com/contents/management-of-acute-appendicitis-in-adults
    - In acute, non-perforated appendicitis then a single dose of antibiotics within 60 minutes of surgery is sufficient
    - In acute, perforated appendicitis then patients should start broad spectrum Gram negative plus anaerobic coverage on diagnosis.
    - Non-operative approach. Exact group to use this in has not been well defined. High risk of failure. May be most appropriate if significant operative risk or sub-acute presentation.
- Gorter, RR; Eker, HH; Gorter-Stam, MAW; Abis, GSA; Acharya, A; Ankersmit, M; Antoniou, SA; Arolfo, S; Babic, B; Boni, L; Bruntink, M; van Dam, DJ; Defoort, B; Deijen, CL; DeLacy, FB; Go, PMNYH; Harmsen, AMK; van den Helder, RS; Iordache, F; Ket, JCF; Muysoms, FE; Ozmen, MM; Papoulas, M; Rhodes, M; Straatman, J; Tenhagen, M; Turrado, V; Vereczkei, A; Vilallonga, R; Deelder, JD; Bonjer, J. Diagnosis and management of acute appendicitis. EAES consensus development conference 2015. Surgical Endoscopy. 2016; 30(11): 4668–4690. doi: 10.1007/s00464-016-5245-7
- National Institute for Health and Clinical Excellence Appendicitis Clinical Knowledge Summary: Scenario: Managing suspected appendicitis (July 2020) https://cks.nice.org.uk/topics/appendicitis/management/managing-suspected-appendicitis/
- Di Saverio et al. 2020. Diagnosis and treatment of acute appendicitis: 2020 update of the WSES Jerusalem guidelines. World Journal of Emergency Surgery (2020) 15:27
- Hansson J., Korner U. Khorram-Manesh A. et al. 2009. Randomized clinical trial of antibiotic therapy versus appendicectomy as primary treatment of acute appendicitis in unselected patients. Br J Surg. 2009; 96: 473-481
- Styrud J., Eriksson S. Nilsson I. et al. 2006. Appendectomy versus antibiotic treatment in acute appendicitis. a prospective multicenter randomized controlled trial. World J Surg. 2006; 30: 1033-1037
- Turhan A.N., Kapan S. Kutukcu E. et al. Comparison of operative and non operative management of acute appendicitis. 2009. Ulus Travma Acil Cerrahi Derg [Turkish J Trauma Emerg Surg]. 2009; 15: 459-462
- Vons C., Barry C. Maitre S. et al. 2011. Amoxicillin plus clavulanic acid versus appendicectomy for treatment of acute uncomplicated appendicitis: an open-label, non-inferiority, randomised controlled trial. Lancet. 2011; 377: 1573-1579
- Eriksson S., Granstrom L. 1995. Randomized controlled trial of appendicectomy versus antibiotic therapy for acute appendicitis. Br J Surg. 1995; 82: 166-169
- Talan DA, Saltzman DJ, Mower WR et al. Antibiotics-First Versus Surgery for Appendicitis: A US Pilot Randomized Controlled Trial Allowing Outpatient Antibiotic Management. 2017. Ann Emerg Med. 2017 Jul; 70(1): 1–11.e9.
- Findlay JM, el Kafsi J, Hammer C et al. 2016. Nonoperative Management of Appendicitis in Adults: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. J Am Coll Surg 2016 Dec;223(6):814-824.e2
- Andersen BR, Kallehave FL, Andersen HK. 2005. Antibiotics versus placebo for prevention of postoperative infection after appendicectomy. Cochrane Database Syst Rev. 2005:CD001439..

8. Associated Documentation

8.1. Evidence summary for recommendations in appendix 1 – Adult appendicitis guideline

Clinical Scoring Systems for Acute Appendicitis in Adults

- A clinical diagnosis of possible appendicitis should be based on an integrated assessment of clinical history, examination findings and blood results
- A number of clinical scoring systems have been reported and variably validated to assist or supplement this process. These may be useful, however, should not be used in isolation to make clinical decisions nor replace senior clinical assessment
- Of these, the WSES recommend the AIR and AAS as useful scoring systems to risk stratify patients

Non-operative Management in Adults (NOTA)

- Recommendation
  - NOTA should be discussed with patients with a clear clinical and radiological diagnosis of uncomplicated appendicitis as a treatment option, with the information and caveats above, and balanced against the risks and benefits of surgery (which remains the gold standard treatment for acute appendicitis)
  - Imaging is advised to make this diagnosis (excluding appendicolith or complication), and additional risks and benefits of this should be discussed with the patient (versus, for example, diagnostic laparoscopy)
  - Patients treated with NOTA should undergo close serial clinical and biochemical review to identify failure to improve early
  - NOTA should not be considered routinely in pregnancy
  - Consideration of subsequent colonic imaging (a colonoscopy as first line) should be considered, particularly in patients aged ≥ 40
- Supporting Evidence
  - There have been 7 RCTs including 1800 patients comparing NOTA to appendicectomy, in addition to a number of non-randomized prospective and retrospective cohort studies (3-9)
  - These RCTs include may areas of potential selection, diagnostic and performance bias
  - These include major differences in inclusion, exclusion and diagnostic criteria; differences in antibiotics and duration therapy; major variability in surgical approach (open procedures being performed in the majority of cases); variable follow up, and variable complication reporting and qualification
  - Meta-analysis of these RCTs (11) suggests that (for selected patients meeting their inclusion criteria), NOTA works in the short term (index admission) for 90% of patients, but fails in the medium term (1 year) in a

quarter of patients, and in the long term (5 years) in nearly a half. The risk thereafter is unknown

- Patients in whom NOTA fails in hospital are much more likely to have complicated appendicitis
- Overall there do not seem to be any clear differences in outcome other than a probable reduction in pain for patients not undergoing surgery.
- There is also a small but important risk of missing an underlying malignancy in 0.5% of patients (even with a normal CT).
- Overall, exclusion criteria in these RCTs effectively amounted to patients without radiological or clinical evidence of a perforation or abscess (with the caveat that clinical evidence is subjective). Beyond this, some trials excluded patients over the age of 50/60/70, female patients, patients with significant comorbidities, patients with a faecolith on imaging, or an appendix greater than 11mm or 15mm in diameter
- Almost but one RCT relied upon imaging confirmation; one required an ultrasound scan, two an ultrasound or CT, and the rest CT.
- Overall, the evidence for NOTA based on these RCTs was qualified using the GRADE classification as 'low' or 'very low' (11)
- This is supported by NICE and the WSES (World Society of Emergency Surgery):
    - NICE - 'After negative imaging, a non-operative management strategy with intravenous fluids and antibiotics can be a safe and effective approach in selected patients with uncomplicated acute appendicitis' (1)
    - WSES : 'We recommend discussing [NOTA] with antibiotics as a safe alternative to surgery in selected patients with uncomplicated acute appendicitis and absence of appendicolith, advising of the possibility of failure and misdiagnosing complicated appendicitis' (2)
    - WSES : 'We suggest against treating acute appendicitis non-operatively during pregnancy until further high-level evidence is available' (2)
    - WSES : 'We suggest both colonic screening with colonscopy and interval full-dose contrast-enhanced CT scan for patients with appendicitis treated nonoperatively if ≥ 40 years old'

Operative Management in Adults

- Recommendation
    - Laparoscopic appendicectomy is the preferred surgical approach in adults
    - Laparoscopic appendicectomy is the preferred approach in children overall, when appropriate expertise and equipment are available
    - Surgery can be safely delayed for up to 24 hours if required for patients with uncomplicated acute appendicitis

- Supporting evidence
  - Both NICE and the WSES recommend laparoscopic appendectomy as the preferred approach
  - NICE recommends appendicectomy as the gold standard treatment for acute appendicitis, and diagnostic laparoscopy for patients with progressive/persistent pain and an unclear diagnosis
  - The WSES recommends that surgery can be safely delayed for up to 24 hours for patients with uncomplicated acute appendicitis

Antibiotic Therapy

Pre-operatively

- Recommendations
  - Patients should receive broad spectrum antibiotics once a diagnosis and management plan has been made
  - Give standard pre-operative antibiotics before surgery, unless less than 1 hour since administration of treatment dose antibiotics on ward
- Supporting Evidence
  - Cochrane review of pre-operative antibiotic therapy (11)
  - WSES guidelines (2)

Post-operatively

- Recommendations
  - Routine antibiotics are not required post-operatively unless localised or generalised peritoneal pus / faecal contamination (complicated appendicitis)
  - If antibiotics are indicated then a course of up to 3-5 days should be prescribed.
  - Continue intravenous gentamicin and metronidazole until
    - Bowels functioning (reliable absorption)
    - Inflammatory markers <75% of peak level
    - Afebrile
  - At this point complete the planned course with Metronidazole 400mg TDS PO plus Trimethoprim 200mg BD PO
    - If patients need more than 5 days of antibiotics then discuss with microbiology and consider imaging
- Supporting Evidence
  - WSES guidelines
  - In patients with complicated acute appendicitis, postoperative broad-spectrum antibiotics are suggested, especially if complete source control has not been achieved. For adult patients deemed to require them, discontinuation of antibiotics after 24 h seems safe and is associated with shorter length of hospital stay and lower costs. In patients with intra-abdominal infections who had undergone an adequate source control, the outcomes after fixed-duration antibiotic therapy

(approximately 3–5 days) are similar to those after a longer course of antibiotics.

- We recommend against prolonging antibiotics longer than 3–5 days postoperatively in case of complicated appendicitis with adequate source control

8.2. Evidence summary for recommendations in appendix 2 – Paediatric appendicitis guideline

Clinical Scoring Systems for Acute Appendicitis in Children

- Recommendation
  - Clinical diagnosis in children may be considerably more challenging
  - The evidence for clinical scoring systems is less robust than for adults; therefore, whilst these may be useful they should not be used in isolation nor replace senior clinical assessment
- Supporting evidence
  - Of these, the WSES recommend the Alvarado Score and Paediatric Appendicitis Score as the most useful scores, but recommend **against** using these in isolation.

Non-operative Management in Children

- Recommendation
  - Appendicectomy remains the gold standard treatment for appendicitis
  - Unless an appendicolith can be excluded on imaging, surgery should be preferred
  - If an appendicolith has been definitively excluded on imaging, NOTA can be discussed with patients/parents as an alternative to surgery, with the below information and caveats
- Supporting evidence
  - The evidence for NOTA in children is considerably less than adults, with just one RCT published to date, and largely consists of non-randomized and retrospective comparisons
  - Overall, meta-analyses of non-randomized studies suggest NOTA has reduced efficacy, an increased readmission rate with comparable rates of complications (2).
  - However, patients with appendicoliths have considerably higher rates of failure and complications
  - In addition, the long term failure rates and implications are unknown
  - The WSES recommend ‘discussing [NOTA] with antibiotics as an alternative to surgery in children with uncomplicated acute appendicitis in the absence of an appendicolith, advising of the possibility of failure and misdiagnosing complicated appendicitis

Operative Management in Children

- Recommendation
  - Laparoscopic appendicectomy is the preferred approach in children overall, when appropriate expertise and equipment are available
  - Surgery can be safely delayed for up to 24 hours if required for patients with uncomplicated acute appendicitis

- Supporting evidence
  - The WSES recommends that 'laparoscopic appendectomy should be preferred over open appendectomy in children where laparoscopic equipment and expertise are available'
  - The WSES recommends that surgery can be safely delayed for up to 24 hours for patients with uncomplicated acute appendicitis

Antibiotic Therapy

Pre-operatively

- Recommendations
  - Patients should receive broad spectrum antibiotics once a diagnosis and management plan has been made
  - Give standard pre-operative antibiotics before surgery, unless less than 1 hour since administration of treatment dose antibiotics on ward
- Supporting Evidence
  - Cochrane review of pre-operative antibiotic therapy (11)
  - WSES guidelines (2)

Post-operatively

- Recommendations
  - Routine antibiotics are not required post-operatively unless localised or generalised peritoneal pus / faecal contamination (complicated appendicitis)
  - If antibiotics are indicated then a course of up to 3-5 days should be prescribed.
  - Continue intravenous gentamicin and metronidazole until
    - Bowels functioning (reliable absorption)
    - Inflammatory markers <75% of peak level
    - Afebrile
  - At this point complete the planned course with oral switch Metronidazole plus Trimethoprim
    - If patients need more than 5 days of antibiotics then discuss with microbiology and consider imaging

- Supporting Evidence
  - WSES guidelines
  - In patients with complicated acute appendicitis, postoperative broad-spectrum antibiotics are suggested, especially if complete source control has not been achieved. For adult patients deemed to require them, discontinuation of antibiotics after 24 h seems safe and is associated with shorter length of hospital stay and lower costs. In patients with intra-abdominal infections who had undergone an adequate source control, the outcomes after fixed-duration antibiotic therapy (approximately 3–5 days) are similar to those after a longer course of antibiotics.
  - We recommend against prolonging antibiotics longer than 3–5 days postoperatively in case of complicated appendicitis with adequate source control

References

1. National Institute for Health and Clinical Excellence Appendicitis Clinical Knowledge Summary: Scenario: Managing suspected appendicitis (July 2020) https://cks.nice.org.uk/topics/appendicitis/management/managing-suspected-appendicitis/
2. Diagnosis and treatment of acute appendicitis: 2020 update of the WSES Jerusalem guidelines. Di Saverio et al. World Journal of Emergency Surgery (2020) 15:27
3. Salminen P., Paajanen H. Rautio T. et al. Antibiotic therapy vs appendectomy for treatment of uncomplicated acute appendicitis: The APPAC randomized clinical trial. JAMA. 2015; 313: 2340-2348
4. Hansson J., Korner U. Khorram-Manesh A. et al. Randomized clinical trial of antibiotic therapy versus appendicectomy as primary treatment of acute appendicitis in unselected patients. Br J Surg. 2009; 96: 473-481
5. Styrud J., Eriksson S. Nilsson I. et al. Appendectomy versus antibiotic treatment in acute appendicitis. a prospective multicenter randomized controlled trial. World J Surg. 2006; 30: 1033-1037
6. Turhan A.N., Kapan S. Kutukcu E. et al. Comparison of operative and non operative management of acute appendicitis. Ulus Travma Acil Cerrahi Derg [Turkish J Trauma Emerg Surg]. 2009; 15: 459-462
7. Vons C., Barry C. Maitre S. et al. Amoxicillin plus clavulanic acid versus appendicectomy for treatment of acute uncomplicated appendicitis: an open-label, non-inferiority, randomised controlled trial. Lancet. 2011; 377: 1573-1579
8. Eriksson S., Granstrom L. Randomized controlled trial of appendicectomy versus antibiotic therapy for acute appendicitis. Br J Surg. 1995; 82: 166-169
9. Talan DA, Saltzman DJ, Mower WR et al. Antibiotics-First Versus Surgery for Appendicitis: A US Pilot Randomized Controlled Trial Allowing Outpatient Antibiotic Management. Ann Emerg Med. 2017 Jul; 70(1): 1–11.e9.

10. Findlay JM, el Kafsi J, Hammer C et al. Nonoperative Management of Appendicitis in Adults: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. J Am Coll Surg 2016 Dec;223(6):814-824.e2
11. Andersen BR, Kallehave FL, Andersen HK. Antibiotics versus placebo for prevention of postoperative infection after appendicectomy. Cochrane Database Syst Rev. 2005:CD001439.

8.3. Trust documentation

- Incident reporting policy
- Antibiotic guidelines for management of severe sepsis and septic shock
- Antibiotic prescribing policy
- Penicillin allergy policy
- Antibiotic guidelines for surgical prophylaxis

9. Appendix 1 – Adult Management of Appendicitis

9.1. Name of guideline on app

Appendicitis and other intra-abdominal infection eg. perforation

9.2. Location on app

Secondary Care
Adult treatment
Gastrointestinal
Appendicitis and other intra-abdominal infection eg. perforation

9.3. Header

Classical presentation: peri-umbilical or epigastric pain that may be colicky, worsens during the first 24 hours (becoming constant and sharp) and migrates to the right iliac fossa (RIF). Always requires surgical and radiological opinion.

9.4. Diagnosis and Differentials

- On examination:
  - Abdominal distension
  - Features of local or general peritonitis: guarding and rebound tenderness or percussion tenderness
  - An abdominal mass may suggest an appendix mass or abscess.
  - Tenderness on percussion, guarding, and rebound tenderness at the RIF. Usually most tender at **McBurney's point**, which lies two-thirds of the way along a line drawn from the umbilicus to the anterior superior iliac spine.
  - Less common peritoneal signs include:
    - **Rovsing's sign** — palpation of the left lower quadrant increases the pain felt in the right lower quadrant.

        - **Psoas sign** — extending the right thigh with the person in the left lateral position elicits pain in the right lower quadrant.
        - **Obturator sign** — internal rotation of the flexed right thigh elicits pain in the right lower quadrant
    - In some cases where diagnosis is uncertain, it may be appropriate to rule out potential alternative causes by considering:
        - Groin and scrotal examination, or
        - Pelvic examination or requesting a gynaecological review.
- Classic signs:
    - Facial flushing, dry tongue, halitosis, low-grade fever (not more than 38°C), and/or tachycardia.
- Classic symptoms:
    - Anorexia — almost always present.
    - Abdominal pain as described above;
        - Often exacerbated by movement, for example coughing or driving over speed bumps
        - Pain may be partly relieved by omental wrapping
    - Gastrointestinal symptoms
        - Nausea and vomiting.
        - Constipation (or sometimes diarrhoea).
    - Systemic symptoms
        - Fever (often low grade), malaise, anorexia.
- Atypical symptoms include
    - Older adults: pain, abdominal signs and fever may be minimal; often associated with confusion
    - Pregnant women: displacement of the appendix may cause right upper quadrant or right flank pain.
    - A retrocaecal/retrocol appendix may cause right loin pain, without signs of peritoneal inflammation
    - A pelvic appendix may cause suprapubic pain and urinary symptoms with abnormal urinalysis,
    - An abnormally long appendix with tip inflammation may cause pain in outside the right lower quadrant.
- Making the diagnosis of acute appendicitis
    - A clinical diagnosis of possible appendicitis should be based on an integrated assessment of clinical history, examination findings and blood results
    - Clinical scoring systems may be of use in addition to senior clinician review when assessing suspected acute appendicitis, but should not be used in isolation to make clinical decisions nor replace senior clinical assessment:
        - AAS https://www.appendicitisscore.com/

        - AIR scoring system https://www.mdcalc.com/appendicitis-inflammatory-response-air-score
    - Imaging required if definitive diagnosis is unclear following clinical assessment. See "always remember to…" for requesting details.

9.5. Always remember to…

- Low threshold for admitting the following groups:
    - Pregnant patients
    - Elderly
    - High risk of complications
    - Clinician or patient feel uneasy about managing a deterioration safely in the community
    - Previous non-operative treatment of appendicitis
- Pregnancy test all post-pubertal / pre-menopusal females (unless undergone hysterectomy) –blood test beta HCG if needs imaging
- Microbiology – before first dose of antibiotics:
    - Blood cultures x2 sets if rigors
    - Stool sample if diarrhoea
    - Urine only if unclear focus of infection or complaining of urinary symptoms
        - Dipstick to detect leukocytes and nitrites for non-pregnant women 16-60, **do not dipstick any other specimens**.
- **Fast until surgical review. Patients may require IVI for hydration, or according to surgeon preference can be allowed clear fluids only.**
- Radiological requests mandatory where non-operative management is proposed OR if diagnostic uncertainty due to surgeon concern re: additional other abdominal pathology:
    - Males: CT scan
    - Females: Ultrasound scan first for ovarian/uterine pathology/atypical pregnancy
        - Request follow-up CT scan if NAD on ultrasound and pregnancy test negative

9.6. Watch Out For (red flags)…[open]

- Peritonitis

9.7. Surgical Management: Pre- and peri- operative antibiotic management of diagnosed appendicitis [open]

- Give antibiotics on diagnosis of appendicitis

    Gentamicin IV [link] as per protocol

    plus

Metronidazole 500mg TDS IV

Consider adding **Amoxicillin 1g QDS IV** if severe infection

- Give standard pre-operative antibiotics before surgery, unless less than 1 hour since administration of treatment dose antibiotics on ward.
- Laparoscopic appendicectomy is the preferred surgical approach in adults
- Surgery can be safely delayed for up to 24 hours if required for patients with uncomplicated acute appendicitis

9.8. Antibiotics in patients undergoing an exploratory procedure when diagnosis uncertain:

- Do not give antibiotics pre-operatively
- If appendicitis is diagnosed during surgery and prophylactic antibiotics have not been given, then give treatment dose antibiotics in theatre

9.9. Post operative antibiotic management [closed]

- Do not give antibiotics post-operatively unless perforation with soiling, or peritoneal contents are turbid/smelly intra-operatively
- If antibiotics are indicated then continue IV gentamicin and metronidazole until
  - Bowels functioning
  - Inflammatory markers below 75% of peak level
  - Afebrile
  - Off inotropes
- Complete planned course with oral switch

  Metronidazole 400mg TDS PO

  plus

  Trimethoprim 200mg BD PO

- Usually antibiotics continue 3-5 days.
- If patients need more than 5 days of antibiotics then discuss with microbiology and consider imaging

9.10. Non-Operative (Conservative) management (NOTA) [closed]

- There is some evidence that conservative management can be effective in carefully selected cases. **For full discussion of evidence and recommendations, see BOB guideline "associated documentation" section.**
  - Clear clinical diagnosis of appendicitis
  - Uncomplicated appendicitis, confirmed by radiological imaging

- The following patients are excluded from the NOTA pathway – surgical intervention required:

- Previous NOTA
- Pregnant patients
- Findings on imaging:
  - Abscess
  - Faecolith / appendicolith
  - Appendix diameter >11mm
  - Rupture / peritonitis
  - Tumour / mass
  - Other pathology requiring intervention

- Patients who qualify for treatment on the NOTA pathway should be informed of the following:
  - Successful NOTA in patients should be followed up with post-treatment colonoscopy ± contrast CT scan to rule out other bowel pathology, particularly if ≥40 years old
  - Short-term 90% efficacy, 25% recurrence at 1 year, 50% recurrence at 5 years. No data for longer-term.
  - Failure of treatment in-hospital much more likely to be complicated appendicitis.
  - Recurrence of symptoms after attempting conservative treatment is an indication for surgical management.
  - No clear differences in outcome other than probable reduction in pain for patients not undergoing surgery.
  - 0.5% patients have underlying malignancy which is missed, as CT scan appears normal

Give standard IV antibiotics initially, and if not for surgery, switch to oral at 48 hours:

Metronidazole 400mg TDS PO

plus

Trimethoprim 200mg BD PO

Total course length usually 7 days (including IVs first)

9.11. If no better…

- Consider alternative diagnosis
- Intra-abdominal collection – seek radiological opinion
- Consider adding amoxicillin for *Enterococcal* cover if not already done
- If on NOTA pathway – immediate consideration for surgical intervention as high risk for complications

9.12. General Interest and other points

Diagnostic signs:

- Pregnancy — Right lower quadrant pain and tenderness dominate in the first trimester. In the latter part of pregnancy, right upper quadrant or right flank pain may occur.
- A retrocaecal/retrocolic appendix may present with right loin pain and tenderness and a positive psoas test. Muscular rigidity and tenderness to deep palpation are often absent because of protection from the overlying caecum.
- A pre-ileal and post-ileal appendix may present with vomiting and diarrhoea.
- A subcaecal and pelvic appendix may present with suprapubic pain and urinary frequency; diarrhoea and tenesmus may be present due to rectal irritation; abdominal tenderness may be lacking, but rectal or vaginal tenderness may be present on the right side; microscopic haematuria and leukocytes may be present on urinalysis.
- A long appendix with tip inflammation in the left lower quadrant may cause pain in that region.

9.13. Other relevant guidelines

9.14. Organism and sensitivity data [closed]

9.15. Version control [closed]

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10. Appendix 2 – Paediatric Management of Appendicitis

10.1. Name of guideline on app

Appendicitis and other intra-abdominal infection eg. perforation

10.2. Location on app

Secondary Care
Paediatric treatment
Gastrointestinal
Appendicitis and other intra-abdominal infection eg. perforation

10.3. Header

Classical presentation: peri-umbilical or epigastric pain that may be colicky, worsens during the first 24 hours (becoming constant and sharp) and migrates to the right iliac fossa (RIF). Always requires surgical and radiological opinion.

10.4. Diagnosis and Differentials

- On examination:
  - Abdominal distension
  - Features of local or general peritonitis: guarding and rebound tenderness or percussion tenderness

- An abdominal mass may suggest an appendix mass or abscess.
- Tenderness on percussion, guarding, and rebound tenderness at the RIF. Usually most tender at **McBurney's point**, which lies two-thirds of the way along a line drawn from the umbilicus to the anterior superior iliac spine.
- Less common peritoneal signs include:
  - **Rovsing's sign** — palpation of the left lower quadrant increases the pain felt in the right lower quadrant.
  - **Psoas sign** — extending the right thigh with the person in the left lateral position elicits pain in the right lower quadrant.
  - **Obturator sign** — internal rotation of the flexed right thigh elicits pain in the right lower quadrant
- In some cases where diagnosis is uncertain, it may be appropriate to rule out potential alternative causes by considering:
  - Groin and scrotal examination, or
  - Pelvic examination or requesting a gynaecological review.

- Classic signs:
  - Facial flushing, dry tongue, halitosis, low-grade fever (not more than 38°C), and/or tachycardia.
- Classic symptoms:
  - Anorexia — almost always present.
  - Abdominal pain as described above;
    - Often exacerbated by movement, for example coughing or driving over speed bumps
    - Pain may be partly relieved by omental wrapping
  - Gastrointestinal symptoms
    - Nausea and vomiting.
    - Constipation (or sometimes diarrhoea).
  - Systemic symptoms
    - Fever (often low grade), malaise, anorexia.
- Atypical symptoms include
  - Infants and young children: non-specific abdominal pain and anorexia, withdrawn.
  - Pregnant women: displacement of the appendix may cause right upper quadrant or right flank pain.
  - A retrocaecal/retrocol appendix may cause right loin pain, without signs of peritoneal inflammation
  - A pelvic appendix may cause suprapubic pain and urinary symptoms with abnormal urinalysis,
  - An abnormally long appendix with tip inflammation may cause pain in outside the right lower quadrant.
- Making the diagnosis of acute appendicitis

- A clinical diagnosis of possible appendicitis may be considerably more challenging to make in children compared to adults. Diagnosis should be based on an integrated assessment of clinical history, examination findings and blood results
- Clinical scoring systems are less robust for paediatric patients. They may be of use in addition to senior clinician review when assessing suspected acute appendicitis, but should not be used in isolation to make clinical decisions nor replace senior clinical assessment:
  - Paediatric Appendicitis Score, for children ages 3-18. Access on the Internet via MDCalc: https://www.mdcalc.com/pediatric-appendicitis-score-pas
- Imaging required if definitive diagnosis is unclear following clinical assessment. See "always remember to…" for requesting details.

10.5. Always Remember To…

- Low threshold for admitting the following groups:
  - Pregnant patients
  - High risk of complications
  - Clinician or patient feel uneasy about managing a deterioration safely in the community
  - Previous non-operative treatment of appendicitis
- Pregnancy test all post-pubertal females (unless undergone hysterectomy) – blood test beta HCG if needs imaging
- Microbiology – before first dose of antibiotics:
  - Blood cultures x2 sets if rigors
  - Stool sample if diarrhoea
  - Urine only if unclear focus of infection or complaining of urinary symptoms
    - Dipstick to detect leukocytes and nitrites for children over 3 months
    - Send clean-catch specimen for MC+S prior to antibiotics
- **Fast until surgical review. Patients may require IVI for hydration, or according to surgeon preference can be allowed clear fluids only.**
- Radiological requests mandatory where non-operative management is proposed OR if diagnostic uncertainty, to risk-stratify subsequent management
  - Ultrasound preferred initial imaging modality for all children
  - Any imaging should be considered in the light of potential risks and benefits e.g. reduced risk of negative appendectomy vs. radiation exposure, time delay to treatment/surgery
    - Optimal imaging modalities should be considered on an individual basis and discussed: radiology consultant, surgeon, the child (if deemed competent) and the parents/carers should be involved.

10.6. Watch Out For (red flags)

- Peritonitis

10.7. Pre- and Peri-operative antibiotic management

- Give antibiotics on diagnosis of appendicitis

  Gentamicin IV [link] as per paediatric gentamicin protocol

  plus

  Metronidazole (child 2 months – 17 years) 7.5mg/kg (max. per dose = 500mg) TDS IV

  Consider adding **Amoxicillin 30mg/kg (max. per dose = 1g) QDS IV** if severe infection

- Give standard pre-operative antibiotics before surgery, unless less than 1 hour since administration of treatment dose antibiotics on ward.
- Laparoscopic appendicectomy is the preferred approach in children overall, when appropriate expertise and equipment are available
- Surgery can be safely delayed for up to 24 hours if required for patients with uncomplicated acute appendicitis

10.8. Post-operative antibiotic management

- Do not give antibiotics post-operatively unless perforation with soiling, or peritoneal contents are turbid/smelly intra-operatively
- If antibiotics are indicated then continue IV gentamicin and metronidazole until
  - Bowels functioning
  - Inflammatory markers below 75% of peak level
  - Afebrile
  - Off inotropes
- Complete planned course with oral switch

  Metronidazole

  2 months – 11 years 7.5mg/kg (max. per dose 400mg) TDS PO

  12 years and over 400mg TDS PO

  plus

  Trimethoprim

  2 months – 5 months 25mg BD PO

  6 months – 5 years 50mg BD PO

  6 years – 11 years 100mg BD PO

  12 years – 17 years 200mg BD PO

- Usually antibiotics continue 3-5 days.
- If patients need more than 5 days of antibiotics then discuss with microbiology and consider imaging

10.9. Non-antibiotic Treatment of Appendicitis (NOTA)

- **For full discussion of evidence and recommendations, see BOB guideline "associated documentation" section.**
- Appendicectomy remains the gold standard treatment for appendicitis
- Unless an appendicolith can be excluded on imaging, surgery should be preferred
- If an appendicolith has been definitively excluded on imaging, NOTA can be discussed with patients/parents as an alternative to surgery, with the below information and caveats:
  - The evidence for NOTA in children is considerably less than adults, with just one RCT published to date, and largely consists of non-randomized and retrospective comparisons
  - Overall, meta-analyses of non-randomized studies suggest NOTA has reduced efficacy, an increased readmission rate with comparable rates of complications.
  - However, patients with appendicoliths have considerably higher rates of failure and complications
  - In addition, the long term failure rates and implications are unknown
  - The WSES recommend 'discussing [NOTA] with antibiotics as an alternative to surgery in children with uncomplicated acute appendicitis in the absence of an appendicolith, advising of the possibility of failure and misdiagnosing complicated appendicitis

10.10. If No Better

- Consider alternative diagnosis
- Intra-abdominal collection – seek radiological opinion
- Consider adding amoxicillin for *Enterococcal* cover if not already done
- If on NOTA pathway – immediate consideration for surgical intervention as high risk for complications

10.11. General Interest and other points

Diagnostic signs:

- Pregnancy — Right lower quadrant pain and tenderness dominate in the first trimester. In the latter part of pregnancy, right upper quadrant or right flank pain may occur.
- A retrocaecal/retrocolic appendix may present with right loin pain and tenderness and a positive psoas test. Muscular rigidity and tenderness to deep palpation are often absent because of protection from the overlying caecum.
- A pre-ileal and post-ileal appendix may present with vomiting and diarrhoea.

- A subcaecal and pelvic appendix may present with suprapubic pain and urinary frequency; diarrhoea and tenesmus may be present due to rectal irritation; abdominal tenderness may be lacking, but rectal or vaginal tenderness may be present on the right side; microscopic haematuria and leukocytes may be present on urinalysis.
- A long appendix with tip inflammation in the left lower quadrant may cause pain in that region.

10.12. Version Control

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